Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

1 h 15 min · 70 marks · 6–7 questions · ≈ 1 min per mark. Every question ends in a calculation or a 3–6 mark explanation. Most frequent: gas-volume moles (15/16), electrolysis (14), esters (14), titration (11), alcohols (11), metals and alloys (11).

TOP 12 MARK-LOSERS

- Equilibrium: concentrations **remain constant**, not "are equal".
- Yield: effect + reason (exothermic / fewer moles of gas). Le Chatelier alone = ignored.
- Catalyst: **alternative pathway** with lower Ea; no effect on yield (both rates up equally).
- **Chloride ions** lose electrons; **Cu^{2+} ions** gain electrons. Never "chlorine/copper".
- Ionic compounds: **ions** move. Metals: **delocalised electrons** move. Mixing them = 0.
- Metals: layers of atoms/ions slide. No molecules or intermolecular forces.
- ΔH needs a sign. Exothermic: more energy released making bonds than taken in breaking.
- Titres to 2 d.p. (23.80); mean of concordant titres (within 0.20 cm^3) to 2 d.p.
- Universal indicator is rejected for titrations.
- Colours in order: dichromate orange → green; methyl orange yellow → red.
- Displayed formulae show every bond, including O–H.
- Never 1 s.f.; standard form when asked; check cm^3 vs dm^3.

ATOMS AND ISOTOPES

ISOTOPES (2)

Same number of protons, different neutrons. Same chemical properties because same **electron configuration**.

- A_r = Σ(mass × %) ÷ 100, round as asked: Rb 85.6, Cl 35.4, Mg 24.4, Br 80.2.
- Cl_2 with 75% ^{35}Cl: P(70) = 0.5625; P(72) = 0.75 × 0.25 × **2** = 0.375.
- Noble gases: full outer shell, so no need to lose, gain or share electrons.

GROUPS 1 AND 7

GROUP 1 DOWN THE GROUP (3)

Larger atom, outer shell further from nucleus → outer electron less attracted → **outer electron more easily lost**.

GROUP 7 UP THE GROUP (4)

F more reactive: smaller atom, outer shell closer to nucleus → stronger attraction for an incoming electron → **gains an electron more readily**.

- Na + water: fizzing, moves, melts into ball, gets smaller, white trail. K: lilac flame. UI blue/purple (OH^-).
- $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$
- Displacement: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ (orange). Iodine + KCl: no change. Cl > Br > I.
- F pale yellow gas · Cl pale green gas · Br brown liquid · I dark grey solid · At black solid.

AIR AND RUSTING

- % O_2 = used ÷ (syringe + flask) × 100: 73 ÷ 380 = 19%.
- Mg burns: bright white light, white powder (not "precipitate").
- NO_x: $N_2 + O_2$ **from air** at high engine temperature. $2NO + 2CO \rightarrow N_2 + 2CO_2$.
- CO_2: greenhouse gas → climate change. No ozone, no acid rain.
- Rust = hydrated iron(III) oxide; needs water + oxygen.
- Zinc/Mg more reactive, so it **oxidises/corrodes instead of iron** (not "rusts").

BONDING AND CONDUCTIVITY

DEFINITIONS

Covalent bond: attraction between a shared pair of electrons and the **nuclei**.

Metallic bond: attraction between positive ions and delocalised electrons.

Alloy: mixture of a metal with other element(s) (not a compound).

ALLOY HARDER (3)

Pure metal: same-size atoms in regular layers that slide. Alloy: different-sized atoms disrupt the layers → layers cannot slide easily.

CONDUCTIVITY (4–5)

Metal: delocalised electrons move. Ionic solid: ions fixed in a lattice. Molten / aqueous: ions free to move. Covalent (water, diamond): no ions or electrons free to move.

- Giant ionic high mp: strong electrostatic forces between oppositely charged ions, large amount of energy to overcome.
- Bigger molecule, higher bp: stronger intermolecular forces, more energy to overcome.
- HCl in organic solvent: molecules, no H^+, litmus stays blue. In water: H^+ forms, litmus red.

ELECTROLYSIS

Cathode (−): reduction	Anode (+): oxidation
$Cu^{2+} + 2e^- \rightarrow Cu$	$2Cl^- \rightarrow Cl_2 + 2e^-$
$2H^+ + 2e^- \rightarrow H_2$	$2Br^- \rightarrow Br_2 + 2e^-$
$Na^+ + e^- \rightarrow Na$	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$
$Pb^{2+} + 2e^- \rightarrow Pb$	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$
$Al^{3+} + 3e^- \rightarrow Al$	$2O^{2-} \rightarrow O_2 + 4e^-$

- Solution: H_2 at cathode unless metal below H (Cu, Ag). Anode: halogen from halide, else O_2.
- H_2 not Na from brine: Na more reactive than H. H^+ attracted to −, gain electrons, pair up.
- $CuSO_4$: pink-brown solid (not red/orange/precipitate); blue fades as Cu^{2+} discharged.
- Graphite electrodes: conduct, high mp, inert. Al anodes burn in O_2 → replaced.
- Less Cl_2 than expected: some dissolves. Electrons (not ions) flow in wires.

METALS

- Above carbon (K–Al): electrolysis. Below carbon (Zn, Fe, Pb): heat with carbon. Give the reason.
- Ti made under argon: argon inert, so Mg/Ti not oxidised.
- Al uses: cans/foil (malleable), aircraft (low density), cables (conductor), pans (conducts heat). Always pair use + property.

MOLES

FORMULAE

$n = m \div M_r \cdot n(gas) = V(dm^3) \div 24 = V(cm^3) \div 24\,000 \cdot n = c \times V(cm^3) \div 1000 \cdot 1\ t = 10^6\ g$

Example	Answer
16 g Zn → H_2	5.9 dm^3
550 cm^3 H_2 → mass Li (2:1)	0.321 g
23.4 t NaCl → Cl_2	4.8×10^6 dm^3
7.7 g $Mg(NO_3)_2$, 5 gas per 2	3.1 dm^3
50.8 g I_2 → HI at 80%	7.68×10^3 cm^3
1570 cm^3 N_2 → NH_3 in 1.1 dm^3	0.238 mol/dm^3
0.55 kg NaOH × 47 kJ/mol	646 kJ
(242 − 60) ÷ 14 repeat units	13

TITRATION CALCULATION

1. n(known) = c × V ÷ 1000 · 2. use the ratio · 3. c = n ÷ V(dm^3). e.g. 21.5 cm^3 0.600 HNO_3 → 0.0129 → ÷ 2 → ÷ 0.025 = 0.258 mol/dm^3. Skipping the ratio gives 0.516: write it on its own line.

TITRATION PRACTICAL

METHOD (6)

Rinse burette with its solution, fill, no air in jet. Pipette 25.0 cm^3 into conical flask. Few drops indicator. Add from burette, swirling, dropwise near end-point, until colour changes permanently. Record initial and final readings. Repeat to concordant.

- Methyl orange: yellow (alkali) → red (acid). Phenolphthalein: pink → colourless.
- White tile: see colour change. Pipette more precise than measuring cylinder.
- Faults: flask rinsed with alkali; burette rinsed with water; no white tile; beaker not conical flask; bung on burette.

SALTS AND SOLUBILITY

SOLUBLE SALT FROM TITRATION (5)

Repeat without indicator, same volumes → heat until crystals start to form → cool → filter/decant → dry in warm oven or filter paper. To dryness = max 1.

INSOLUBLE SALT (3–5)

Mix solutions → filter → wash with distilled water → dry (warm oven/filter paper). $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$.

- Saturated: max solute dissolved in the solvent at a given temperature.
- Scale by water used ÷ 100. Crystallised = solubility(hot) − solubility(cold), × scale factor.
- Draw read-off lines on the graph; extend the curve before extrapolating.

ENERGETICS

- Exothermic: **heat** energy given out to surroundings. Q = mcΔT, m = solution mass (25 + 22 = 47 g).
- ΔH = Q ÷ moles, with sign: ethanol −18.2 kJ ÷ 0.020 mol = −910 kJ/mol.
- Low results: heat loss, incomplete combustion (soot), evaporation. Polystyrene: insulator.

BOND ENERGIES

ΔH = Σ broken − Σ made. $2H_2 + O_2$: 1370 − 1852 = −482. $H_2 + Cl_2$: 678 − 862 = −184. Unknown bond: solve for x (C–H 394; C–Cl 338).

Esterification ΔH ≈ 0: same bonds (C–O, O–H) broken and formed.

- Diagram: products below reactants (exo); ΔH arrow down from reactants to products; Ea arrow up from reactants to peak. Catalysed: same ends, lower peak.

RATES AND EQUILIBRIUM

TEMPERATURE (4)

Particles gain kinetic energy → more collisions per unit time → more collisions ≥ Ea → more frequent successful collisions → faster.

DYNAMIC EQUILIBRIUM (2)

Forward rate = backward rate; concentrations remain constant.

Change	Yield	Because
T down, exothermic forward	up	forward exothermic
T up, endothermic forward	up	forward endothermic
P up, fewer gas moles on right	up	fewer moles of gas on right

Change	Yield	Because
P, equal gas moles (H_2 + I_2)	none	same moles both sides
catalyst	none	both rates up equally

- Catalyst comparison (5): same volume H_2O_2, same mass catalyst, same T; time; gas volume in fixed time; compare gradients.
- Faster curve: steeper from origin, same final volume (if limiting reactant unchanged).

CRUDE OIL

FRACTIONAL DISTILLATION (4)
Heated → vaporises → vapours rise up column → hotter at bottom → condense at their boiling points at different heights.

- Kerosene: **fuel** for aircraft ("aircraft" alone = 0). Bitumen: roads. Refinery gas: bottled gas.
- Cracking: long chains → shorter; silica/alumina; 600–700 °C.
- Why: surplus long chains, demand for petrol; short alkanes more flammable; alkenes for polymers.
- $C_{12}H_{26} \rightarrow C_9H_{20} + C_3H_6$. Bromine water + alkene: orange → colourless.

ALCOHOLS

	Hydration	Fermentation
conditions	steam, 300 °C, 60–70 atm, H_3PO_4	yeast, 30 °C, no air
source	crude oil (finite)	sugar cane (renewable)
speed / process	fast, continuous	slow, batch
product	pure	impure, needs distilling

- $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \cdot C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$
- No air: O_2 oxidises ethanol to ethanoic acid. Above 40 °C enzymes denature.
- Oxidation: acidified $K_2Cr_2O_7$, orange → green, → ethanoic acid.
- Molecular formula of ethanol: C_2H_6O. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.

ACIDS, ESTERS, POLYMERS

- $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$ (fizzing).
- $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$.
- Ester = alcohol-**yl** + acid-**oate**: ethyl ethanoate; methyl propanoate ($CH_3CH_2COOCH_3$); butyl ethanoate ($C_6H_{12}O_2$).
- Conc. H_2SO_4 catalyst; water bath (flammable); fruity smell.
- Displayed ester: –C(=O)–O– in full (M1) + rest (M2).

CONDENSATION POLYMER
Diol + dicarboxylic acid → polyester + water (2n H_2O). Addition: alkene, only product. Condensation: two functional groups, water also made. Polyesters can be biodegradable.

TESTS

Test	Result
Li^+ / K^+ / Cu^{2+} flame	red / lilac / blue-green
NaOH: Cu^{2+} / Fe^{2+}	blue ppt / green ppt
Cl^-: HNO_3 + $AgNO_3$	white ppt
CO_3^{2-}: acid, gas → limewater	fizz; milky
H_2 lighted splint / O_2 glowing	pop / relights
Cl_2 damp litmus	bleached
water: anhydrous $CuSO_4$	white → blue
pure ethanol	boils at 78 °C exactly

FINAL 60 SECONDS

- Count marks = count points. Link with "so" / "because".
- Name the moving particle. Name the ion, not the element.
- Moles → ratio → units → s.f. → sign.